KRISHNA SATHVIKA GANNI

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SKILLS

- **Programming Languages:** Python, SQL, R, PL/SQL
- **Generative AI & LLMs:** LangChain, LangGraph, Retrieval-Augmented Generation (RAG), LLMs, Prompt Engineering, Embeddings, Vector Databases, AI Agents
- **Machine Learning & Deep Learning:** Scikit-learn, PyTorch, Supervised & Unsupervised Learning, Ensemble Methods, Feature Engineering, Time Series Forecasting, YOLOv8, OpenCV
- **Data Engineering & Databases:** PostgreSQL, BigQuery, Oracle, SQLite, ETL/ELT, Apache Airflow, REST APIs, Change Data Capture (CDC), Slowly Changing Dimensions (SCD Type 2), CTEs, Window Functions, Query Optimization
- **Data Analysis & Statistics:** Pandas, NumPy, EDA, Statistical Modeling, Hypothesis Testing, A/B Testing, Bayesian Inference
- **MLOps & Deployment:** MLflow, Model Monitoring, Data Drift Detection, Model Deployment, FastAPI
- **Visualization & BI:** Tableau, Power BI, Looker Studio, Plotly
- **Tools & Development:** Git, GitHub, GitHub Actions, pytest, Docker

CERTIFICATIONS

- AWS Certified AI Practitioner, Amazon Web Services, 2026
- Python for Data Science, YBI Foundation, 2022

EXPERIENCE

ML/AI Research Assistant, New Jersey Institute of Technology (NJIT), Newark, NJ **Oct 2025 - Present**

- Co-authored a peer-reviewed **IEEE Xplore** publication on machine learning for storm-related energy transmission safeguarding, with the associated poster earning **3rd Place, Best Poster** at the 2025 NJIT Future Energy Transmission Conference.
- Improved AI-based storm damage risk prediction models for power grid resilience using **TensorFlow** and **PyTorch**, achieving a **15% increase in model accuracy** through dataset preparation, optimization, and image augmentation techniques.
- Supported **Smart Health AI research initiatives** across breast cancer, Parkinson's, Alzheimer's, glioma, and healthcare cost prediction by conducting literature reviews, preparing datasets, and exploring **explainable AI (XAI)** approaches.

PROJECTS

Enterprise RAG Knowledge Assistant **2026**

Tech: *Python, FastAPI, LangChain, ChromaDB, FastEmbed, Google Gemini, Streamlit*

- Developed a full-stack **Retrieval-Augmented Generation (RAG)** application that ingests PDF, TXT, and DOCX documents and answers natural-language questions using source-attributed responses grounded in retrieved document context.
- Built a document ingestion pipeline (LangChain loaders, recursive text chunking) and a persistent vector search layer using **FastEmbed** embeddings and **ChromaDB**, exposed through a **FastAPI** REST API with multi-model fallback and a Streamlit dashboard.

AI Data Analyst Agent **2026**

Tech: *Python, LangGraph, Pandas, NumPy, Plotly, FastAPI, Streamlit, Google Gemini*

- Built a **LangGraph**-orchestrated analytics agent that ingests CSV/Excel files and answers natural-language business questions using a 4-node stateful workflow (load, analyze, visualize, explain), deployed live on **Render**.
- Implemented deterministic query classification and **Pandas/NumPy** analysis functions (ranking, trend aggregation, linear forecasting), keeping numerical computation fully separate from LLM output, validated via pytest and **GitHub Actions CI**.

MLOps Model Lifecycle Platform **2026**

Tech: *Python, scikit-learn, MLflow, FastAPI, Evidently, PostgreSQL*

- Built an end-to-end MLOps pipeline for customer churn prediction, training a Random Forest classifier and tracking experiments, metrics, and model versions with **MLflow**.
- Implemented automated data drift detection using **Evidently**, triggering a retraining and re-registration workflow when data distribution shifts, served via FastAPI with **100% test coverage** enforced in GitHub Actions CI.

Computer Vision Quality Inspection System **2026**

Tech: *Python, YOLOv8, PyTorch, OpenCV, FastAPI, Streamlit*

- Built an end-to-end computer vision application using **YOLOv8** for image-based object detection, generating annotated output images with OpenCV.
- Implemented a confidence-based quality-grading system, validated with a dedicated pytest suite, deployed live on **Streamlit Community Cloud**, and containerized with **Docker Compose**.

Clinical Trials Landscape Analyzer **2025**

Tech: *Python, SQL, PostgreSQL, Apache Airflow, Tableau*

- Built an end-to-end ETL pipeline extracting **5,000+ clinical trials** from the ClinicalTrials.gov API into a normalized PostgreSQL database, with 12 SQL analytics views (window functions, PERCENTILE_CONT) powering a **Tableau** dashboard.
- Extended the pipeline with a **Change Data Capture (CDC)** and SCD Type 2 layer, using SHA-256 record hashing and field-level diffing over daily snapshots, orchestrated via **Apache Airflow** to track historical trial changes.

PJM Hourly Energy Demand Forecasting **2025**

Tech: *Python, ARIMA, Prophet, LightGBM, FastAPI, Streamlit*

- Built a time-series forecasting pipeline predicting **168-hour-ahead** electricity demand, engineering lag and calendar features with a leakage-free time-based train/test split.
- Trained and benchmarked ARIMA, Prophet, and LightGBM; the gradient-boosting model achieved a **2.33% MAPE**, outperforming ARIMA (5.83%) and Prophet (7.35%), served via FastAPI and a live-deployed Streamlit dashboard.

A/B Testing and Experimentation Analysis Framework **2025**

Tech: *Python, SciPy, statsmodels, NumPy, Pandas*

- Built a reusable A/B testing framework applied to a real-world mobile game retention experiment (**~90,000 players**), implementing frequentist hypothesis testing (z-test, chi-square, Welch t-test, Mann-Whitney U) and power/sample-size analysis.
- Implemented a **Bayesian** Beta-Binomial comparison (Monte Carlo posterior sampling) and a dual-evidence decision rule combining frequentist and Bayesian results into a deploy/retain/inconclusive recommendation.

Web Analytics Funnel Analysis **2025**

Tech: *SQL, BigQuery, Python, Looker Studio*

- Analyzed **108,285 sessions** of real-world GA4 e-commerce event data in **BigQuery**, writing SQL with CTEs, window functions, and nested UNNEST operations.
- Built a 6-step session-based conversion funnel identifying the largest drop-off point, visualized through an interactive **Looker Studio** dashboard.

EDUCATION

New Jersey Institute of Technology (NJIT), Newark, NJ **Jan 2025 - Dec 2026**
Master of Science in Computer Science GPA: 3.94/4.0

Ramachandra College of Engineering, Eluru, India **Aug 2020 - May 2024**
Bachelor of Technology in Computer Science and Engineering CGPA: 8.25/10.0